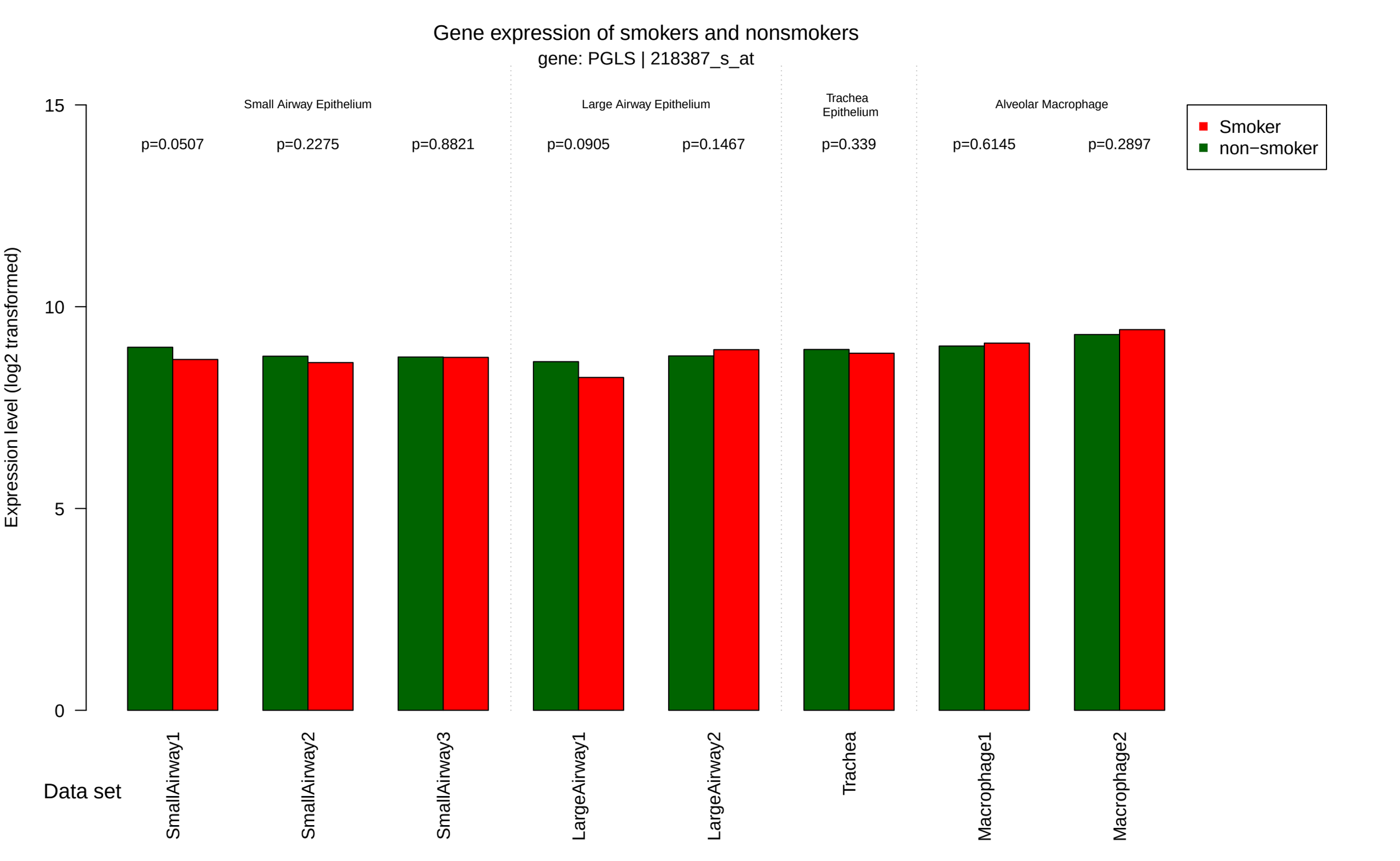
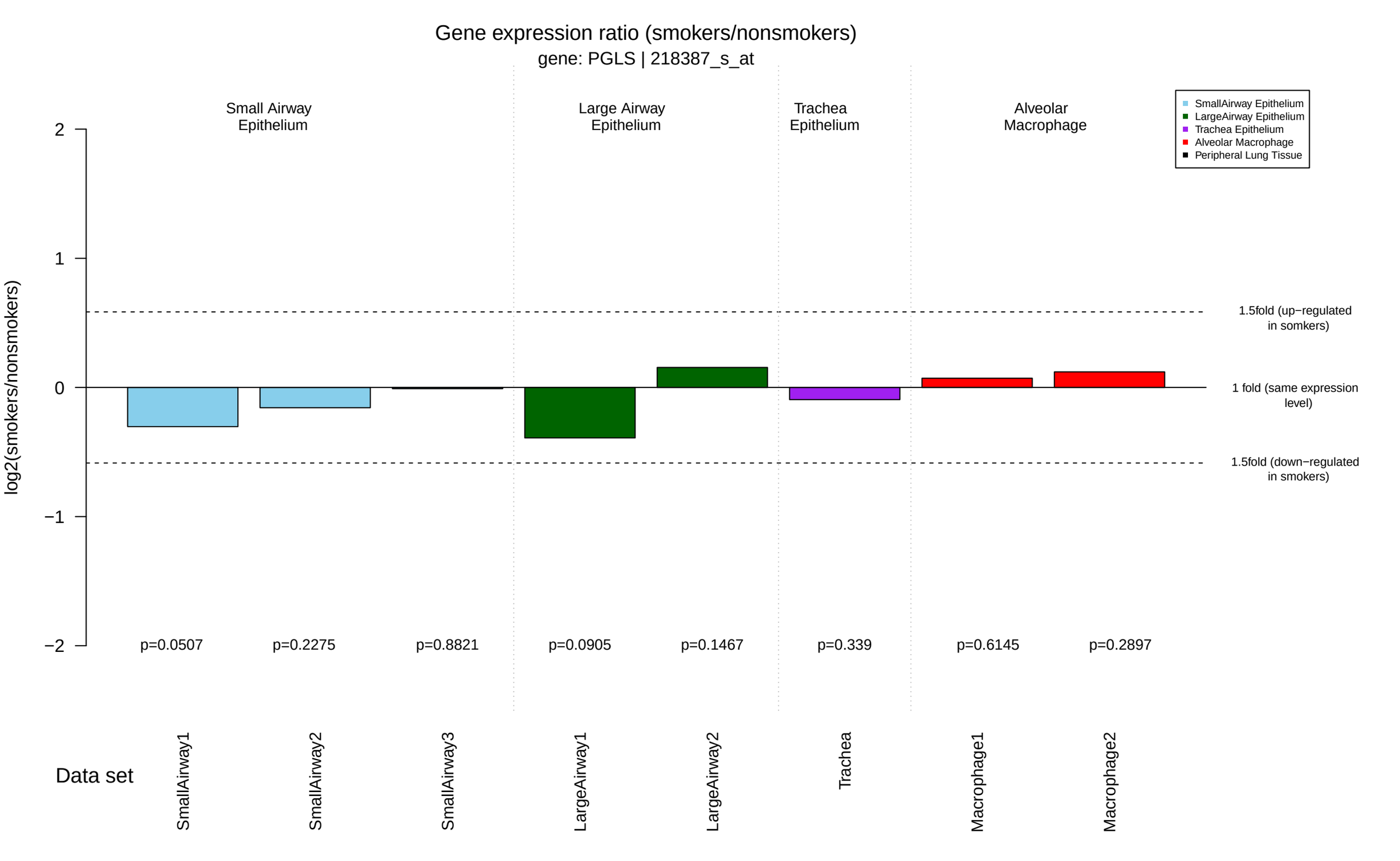
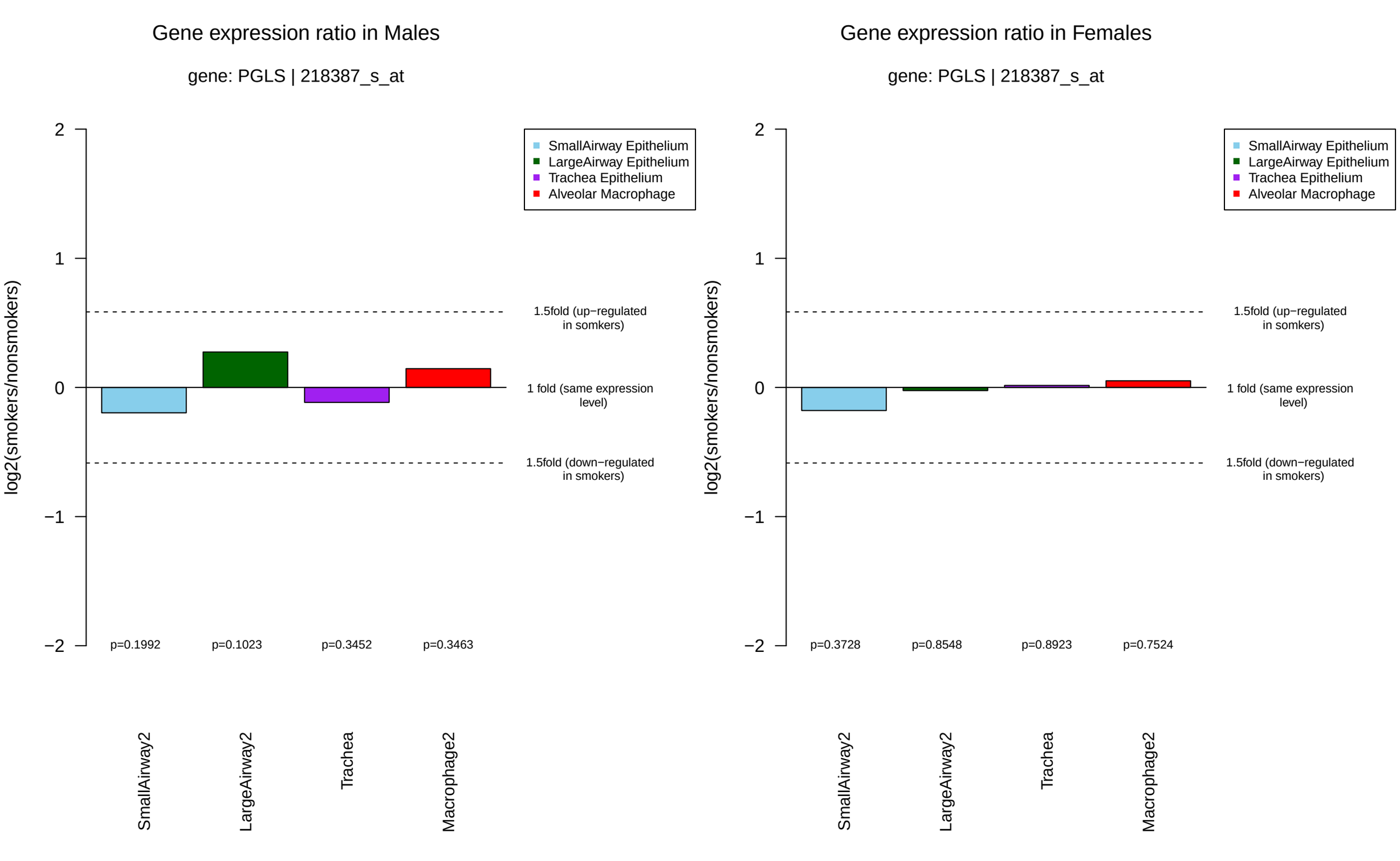
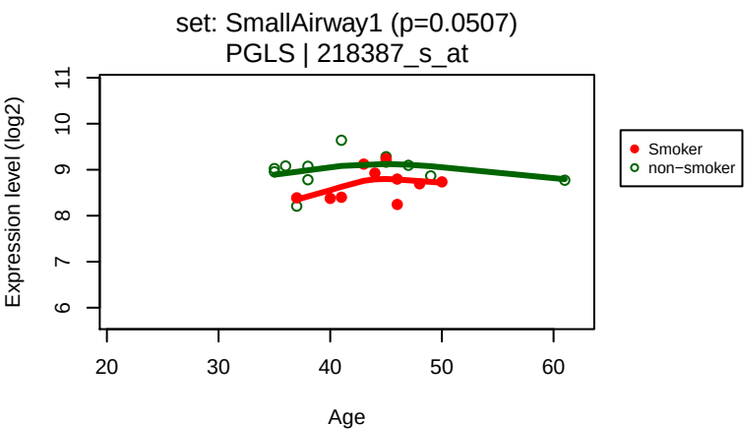
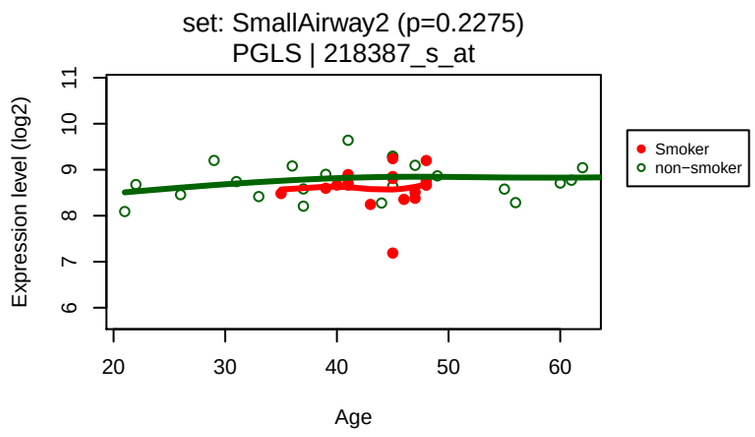
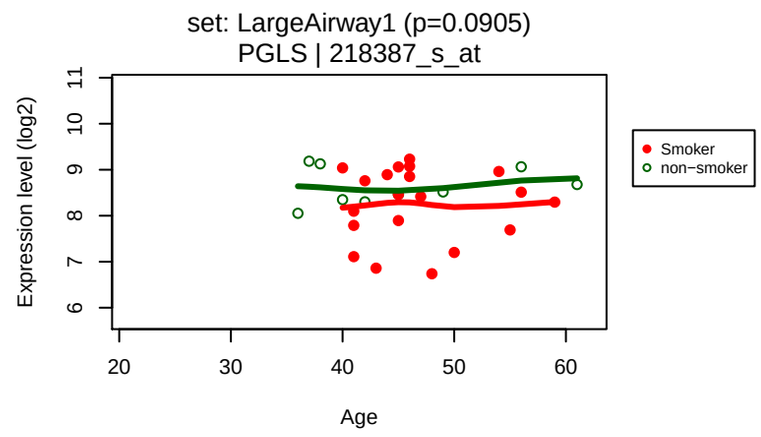

Gene expression of smokers and nonsmokers
gene: PGLS | 218387_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0507
p=0.2275
p=0.8821
p=0.0905
p=0.1467
p=0.339
p=0.6145
p=0.2897
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

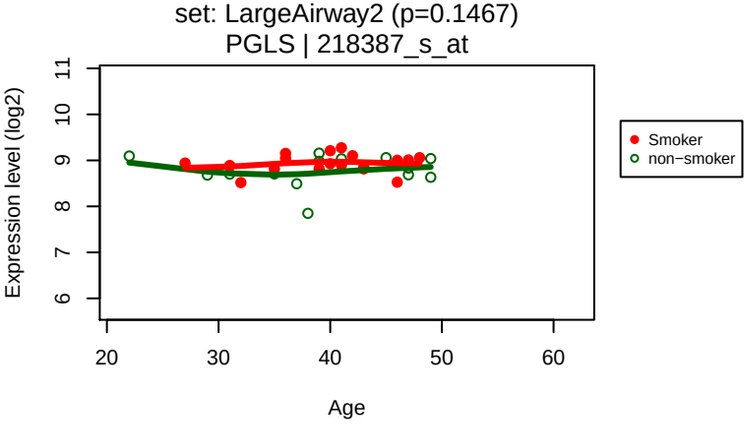

Gene expression ratio (smokers/nonsmokers)
gene: PGLS | 218387_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.2275
p=0.8821
p=0.0905
p=0.1467
p=0.339
p=0.6145
p=0.2897
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

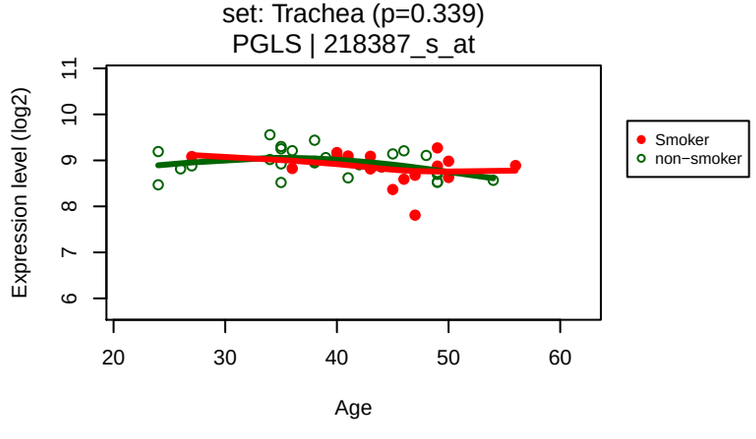

Gene expression ratio in Males
gene: PGLS | 218387_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1992
p=0.1023
p=0.3452
p=0.3463
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PGLS | 218387_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3728
p=0.8548
p=0.8923
p=0.7524
SmallAirway2
LargeAirway2
Trachea
Macrophage2

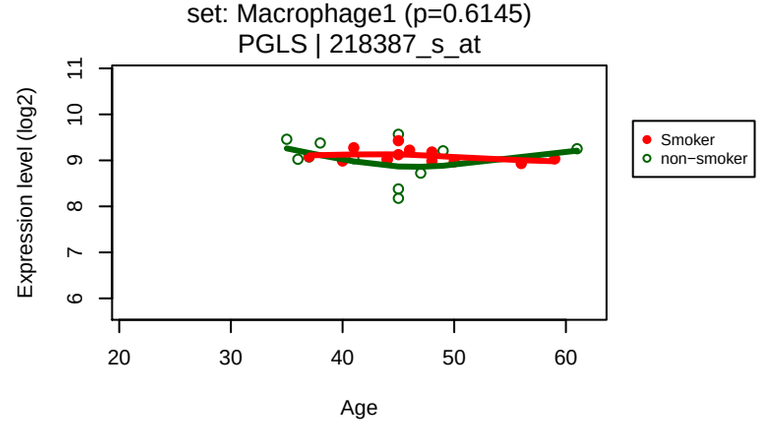

set: SmallAirway1 (p=0.0507)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker

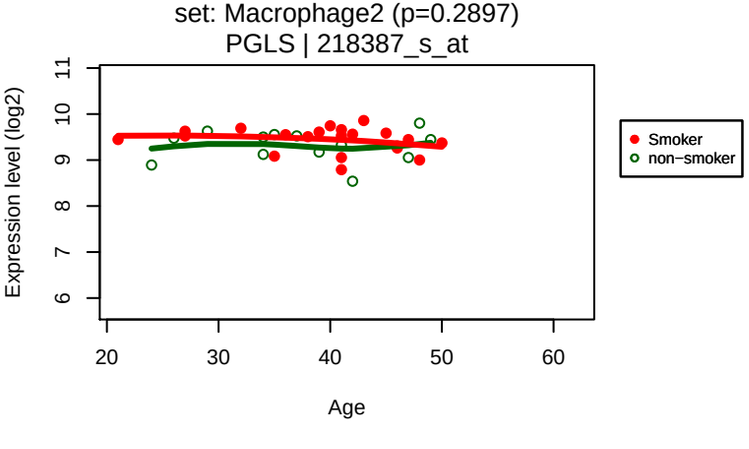

set: SmallAirway2 (p=0.2275)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker

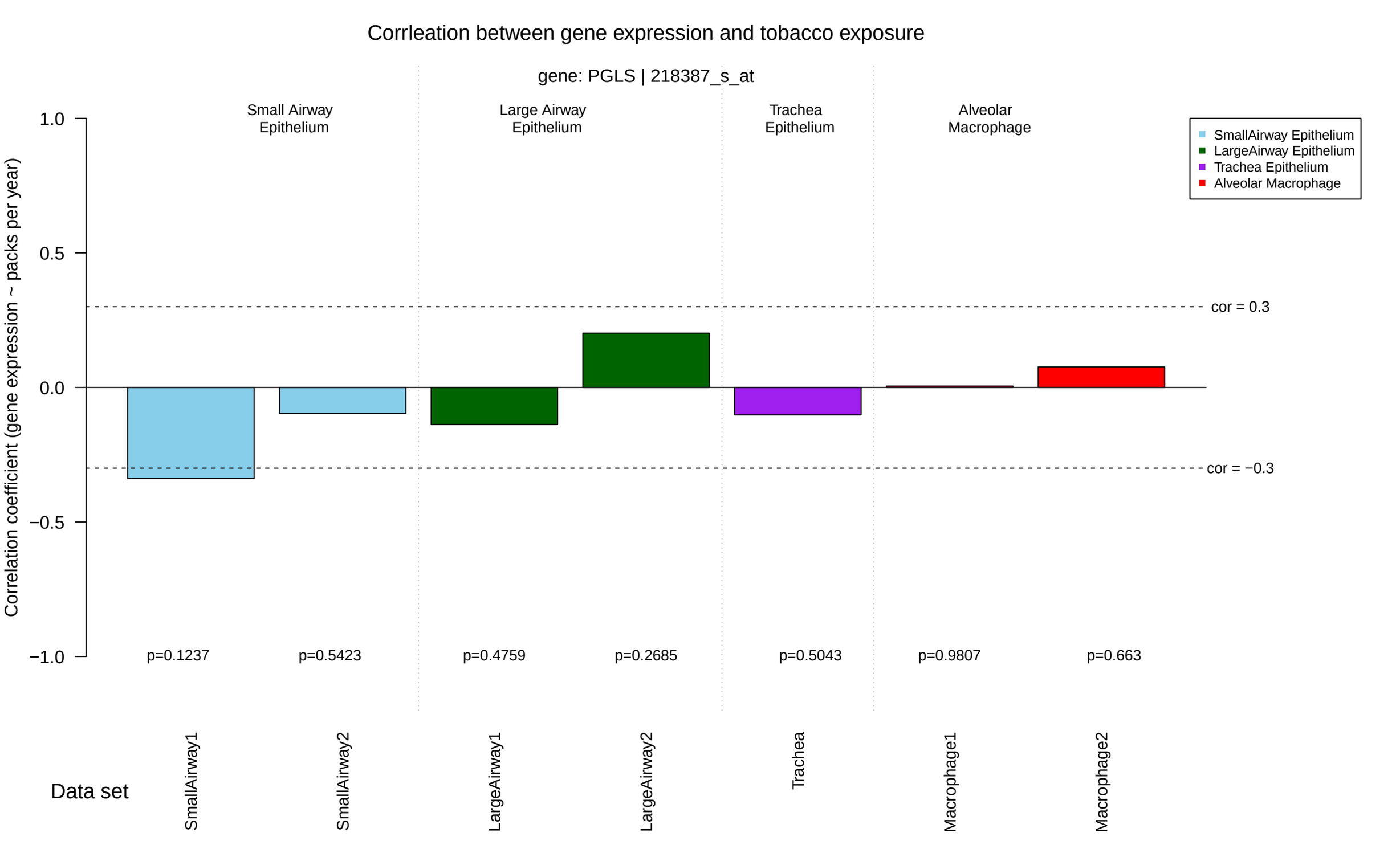

set: LargeAirway1 (p=0.0905)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1467)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.339)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6145)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2897)
PGLS | 218387_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PGLS | 218387_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1237
p=0.5423
p=0.4759
p=0.2685
p=0.5043
p=0.9807
p=0.663
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2